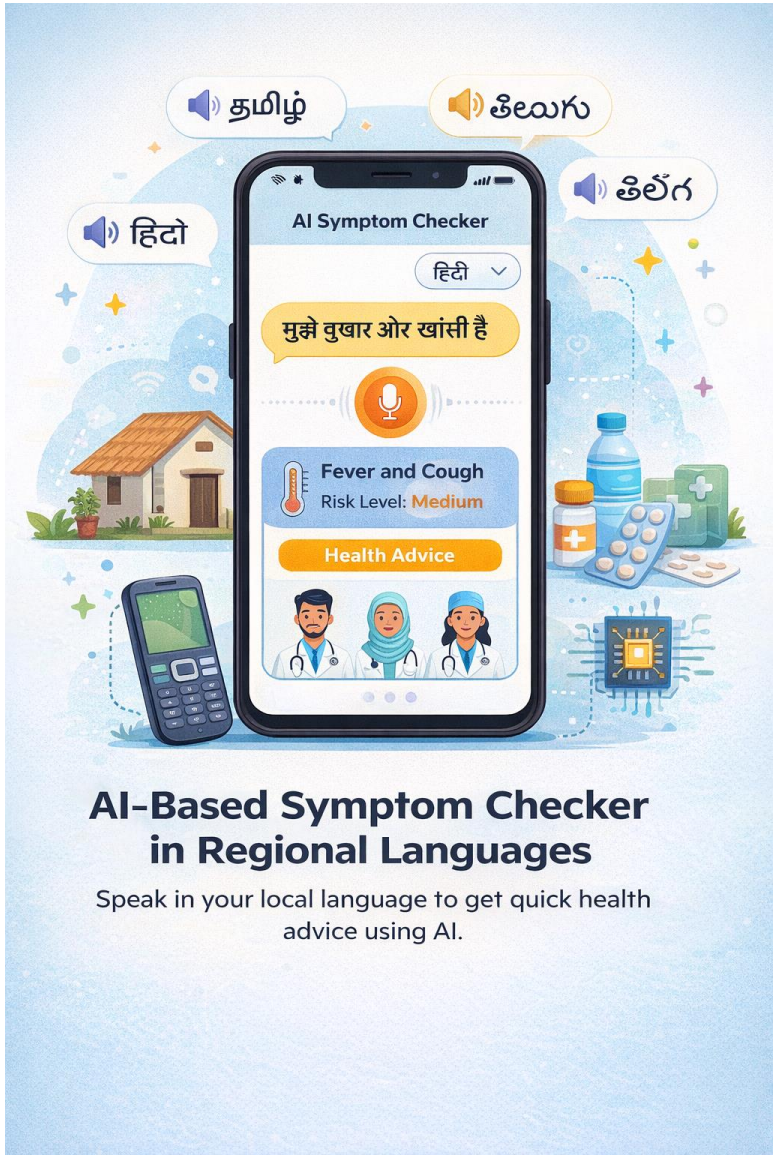
MULTILINGUAL AI SYMPTOM CHECKER

Smart Diagnosis System Using Artificial Intelligence

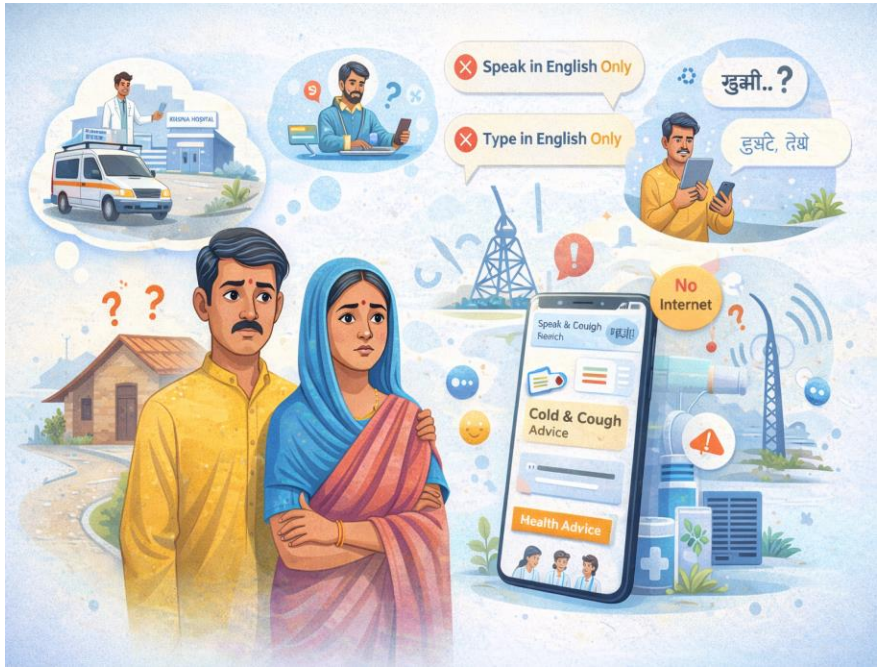
INTRODUCTION

- This project is a smart health tool that uses Artificial Intelligence (AI) to understand user symptoms. It allows people to speak or type their health problems in their regional language like Hindi, Telugu, Tamil, etc. The system processes the input and provides simple health guidance.
- This tool is specially designed for people living in villages and small towns where access to doctors is limited. It helps users get quick information about their health without needing medical knowledge. The system gives a risk level (low, medium, high) instead of a full diagnosis.
- The project also supports voice input, making it easy for people who cannot read or write. It can work in offline mode, which is useful in areas with poor internet connectivity.
- Overall, this project aims to make healthcare support simple, fast, and accessible for everyone.



PROBLEM STATEMENT

- Many people in villages and small towns do not have quick and easy access to doctors or hospitals. They often need to travel long distances for basic medical advice. This causes delay in getting proper treatment.
- Most existing health apps and symptom checkers are only available in English. Many people are not comfortable reading or understanding English, which makes these apps difficult to use.
- In addition, many users prefer speaking in their local language instead of typing. However, most apps do not support voice input in regional languages.
- Another major problem is poor internet connectivity in rural areas. Because of this, online health services do not work properly all the time.
- Due to these challenges, people are unable to get timely and correct health guidance. There is a need for a simple, local-language-based, and offline-friendly solution.



SOLUTION

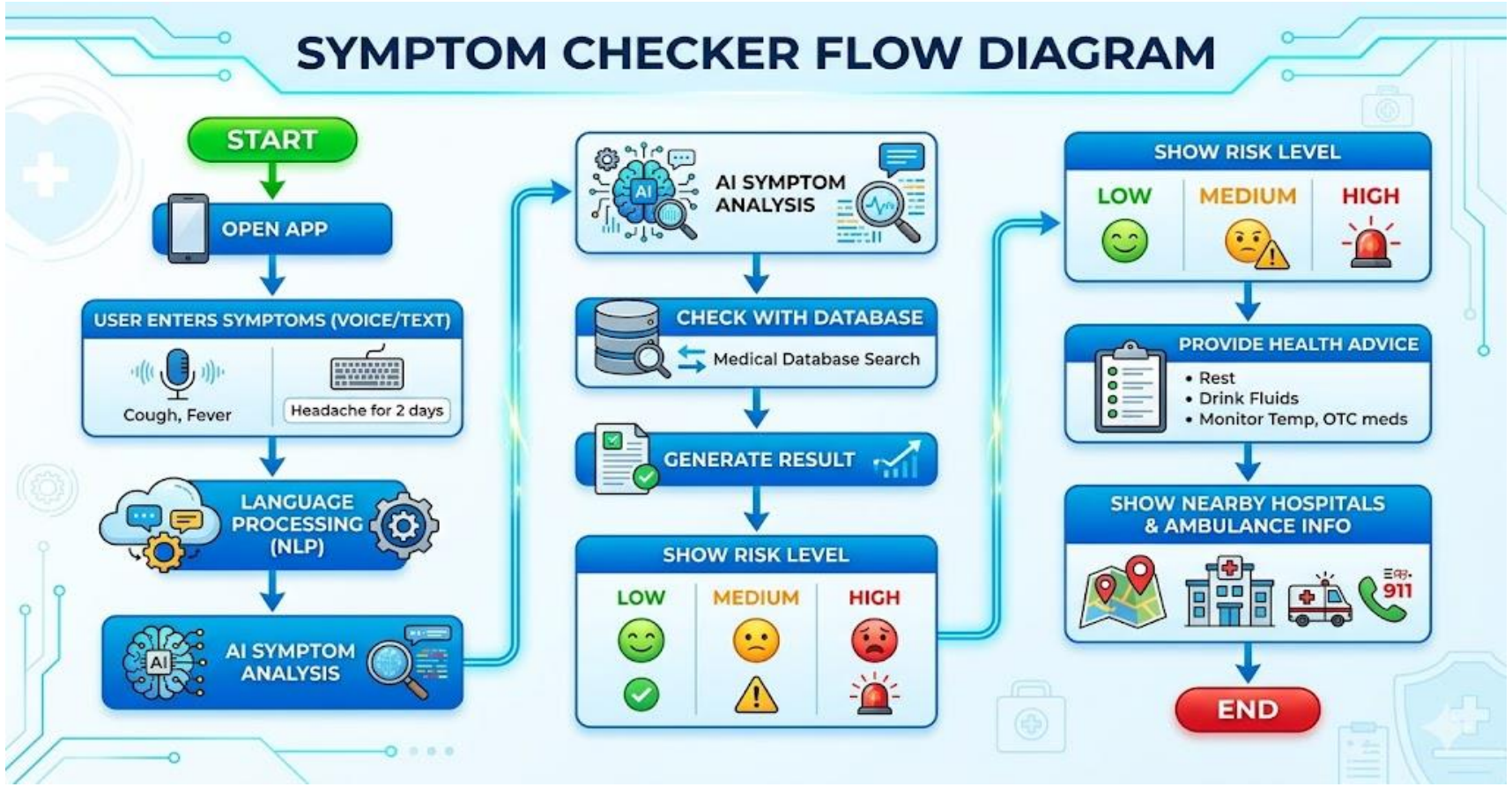
- Many people living in villages and small towns do not have quick access to doctors or hospitals. They often need to travel long distances even for small health problems, which takes time and money.
- Most health applications and symptom checkers are available only in English. Many people are not comfortable reading or understanding English, so they find these apps difficult to use.
- People in rural areas usually prefer speaking in their local language rather than typing. However, many apps do not support voice input or regional languages, which creates a communication gap.
- Another big problem is poor internet connectivity. In many villages, internet connection is slow or not available at all. Because of this, online health services cannot be used properly.
- Many users also do not have much technical knowledge, so complex apps are hard for them to understand and use.
- Due to all these problems, people are not able to get quick and correct health guidance when they need it. This can lead to delays in treatment and sometimes serious health issues.
- Therefore, there is a strong need for a simple, easy-to-use, and language-friendly health solution that can work even without internet.

CORE FEATURE

- The system supports multiple regional languages, allowing users to speak or type in their local language and receive responses in the same language. It uses AI to understand symptoms and provides possible conditions with a simple risk level such as low, medium, or high.
- The application also works in offline mode by using a local database, making it useful in areas with poor internet connectivity. It is designed with a simple and user-friendly interface, including voice input and easy icons, so that even non-technical users can use it without difficulty.
- The system can also help users find nearby hospitals and clinics using location services. In case of emergency, it can suggest the nearest medical centers for quick help. It also provides basic ambulance support information, so users can quickly contact emergency services when needed. This feature helps save time and ensures faster medical assistance in critical situations.

FLOW DIAGRAM

- The flow diagram explains how the system processes user input step by step. The user enters symptoms, and the AI analyzes them using language processing and a database. Finally, the system provides a risk level, health advice, and nearby hospital information.



TECHNOLOGIES USED & HOW IT WORKS

- This project uses technologies like Artificial Intelligence (AI), Natural Language Processing (NLP), Speech Recognition, and Translation APIs to understand user symptoms in different languages. It is developed as a simple mobile application that is easy to use.
- The working of the system is very simple. First, the user opens the app and speaks or types their symptoms in their local language. The system then converts the input into text and analyzes it using AI. After that, it compares the symptoms with a stored database and finds possible health conditions. Finally, the app shows the result with a risk level (low, medium, or high) along with basic health advice.

IMPACT & BENIFITS

- This project can make a big difference in improving healthcare access, especially in rural and remote areas. It helps people understand their health problems quickly without depending only on doctors.
- The use of regional languages makes it easy for users to interact with the system comfortably. Voice input allows even non-literate people to use the application without difficulty.
- The offline feature ensures that the system works even in areas with poor internet connectivity. This makes it more reliable and useful in villages.
- It saves time and money by reducing unnecessary hospital visits. Users can get basic guidance at home and decide when to visit a doctor.
- The feature of finding nearby hospitals and ambulance support helps in emergency situations, ensuring faster medical assistance.
- Overall, this project improves awareness, provides quick help, and makes healthcare more accessible, simple, and user-friendly for everyone.

RESEARCH & REFERENCES

- This project is based on basic research about Artificial Intelligence, healthcare systems, and language processing technologies. Information was collected from various online resources, articles, and educational platforms.
- The concept of AI in healthcare was studied to understand how machines can analyze symptoms and provide guidance. Natural Language Processing (NLP) was researched to support multiple regional languages and voice input.
- References include:
- Google Translate API (for language translation)
- Basic AI and NLP concepts from online tutorials
- Health-related information from general medical websites
- Articles on rural healthcare challenges in India
- This research helped in designing a simple and useful system for real-world healthcare problems.

thank
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